TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	62.0 ± 3.49
Correlation Stability	87.9 ± 1.04
Distribution Consistency	73.6 ± 4.41
Entropy Preservation	67.2 ± 1.17
Mean Z Score Drift	0.5 ± 0.03
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	5.4 ± 1.85
Data Density	100.0 ± 0.00
Coverage Score	87.6 ± 4.10
Feature Accuracy	53.4 ± 1.69
Novelty Score	0.7 ± 0.03

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Dataset Audit: heart_failure_clinical_records.csv

1. Dataset Portrait

Attribute	Value
Rows	299
Columns	13
Numeric features	13
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	60.834	63.159	+2.325	3.8%	OK
anaemia	0.431	0.212	-0.219	50.8%	Drift
creatinine_phosphokinase	581.839	504.071	-77.769	13.4%	Warn
diabetes	0.418	0.232	-0.186	44.4%	Drift
ejection_fraction	38.084	34.313	-3.770	9.9%	OK
high_blood_pressure	0.351	0.141	-0.210	59.7%	Drift
platelets	263358.029	254856.396	-8501.633	3.2%	OK
serum_creatinine	1.394	1.707	+0.313	22.5%	Warn
serum_sodium	136.625	135.879	-0.747	0.5%	OK
sex	0.649	0.424	-0.225	34.6%	Drift
smoking	0.321	0.030	-0.291	90.6%	Drift
time	130.261	69.283	-60.978	46.8%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	62.0 ± 3.49
Correlation Stability (*2)	87.9 ± 1.04
Distribution Consistency (*3)	73.6 ± 4.41
Entropy Preservation (*4)	67.2 ± 1.17
Mean Z Score Drift (*5)	0.5 ± 0.03
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	5.4 ± 1.85

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	87.6 ± 4.10
Feature Accuracy (*10)	53.4 ± 1.69
Novelty Score (*11)	0.7 ± 0.03

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8300000000000001	N/A	0.8233333333333333	-0.00666666666666665	
F1	0.7532401666570763	N/A	0.749209495637223	-0.00403067101985316	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.835	N/A	0.8533333333333333	0.01833333333333333	
F1	0.7547418227770806	N/A	0.7901040947763853	0.03536227199930466	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7849999999999999	N/A	0.7633333333333333	-0.02166666666666678	
F1	0.6802488822308165	N/A	0.656300671438396	-0.023948210792420453	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7499999999999999	N/A	0.7833333333333333	0.03333333333333331	
F1	0.5667171896628354	N/A	0.706318660016139	0.13960147035330372	*

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6333333333333334	N/A	0.5983333333333334	-0.03499999999999996	
F1	N/A	N/A	0.2879656372359410 4	0.28796563723594104	*

Model: AB

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.825	N/A	0.825	-1.1102230246251566e-17	
F1	0.7417404771442427	N/A	0.752263968748623	0.010523491604380297	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8333333333333334	N/A	0.8516666666666668	0.018333333333333333	
F1	0.7399559394144454	N/A	0.7883671541758368	0.048411214761391264	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8066666666666666	N/A	0.7716666666666667	-0.03500000000000001	
F1	0.6937863401621888	N/A	0.7025583613340471	0.008772021171858425	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5533333333333333	N/A	0.5633333333333334	0.010000000000000004	
F1	0.1917473972165231	N/A	0.16418728696552615	-0.02756011025099695	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6133333333333335	N/A	0.5366666666666667	-0.07666666666666666	*
F1	0.007142857142857143	N/A	0.4634402846426805	0.4562974274998234	*

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	75.0	0.0	582.0	0.0	20.0	1.0	265000.0	1.9	130.0	1.0
2	55.0	0.0	7861.0	0.0	38.0	0.0	263358.03	1.1	136.0	1.0
3	65.0	0.0	146.0	0.0	20.0	0.0	162000.0	1.3	129.0	1.0
4	50.0	1.0	111.0	0.0	20.0	0.0	210000.0	1.9	137.0	1.0
5	65.0	1.0	160.0	1.0	20.0	0.0	327000.0	2.7	116.0	0.0
6	90.0	1.0	47.0	0.0	40.0	1.0	204000.0	2.1	132.0	1.0
7	75.0	1.0	246.0	0.0	15.0	0.0	127000.0	1.2	137.0	1.0
8	60.0	1.0	315.0	1.0	60.0	0.0	454000.0	1.1	131.0	1.0
9	65.0	0.0	157.0	0.0	65.0	0.0	263358.03	1.5	138.0	0.0
10	80.0	1.0	123.0	0.0	35.0	1.0	388000.0	9.4	133.0	1.0
11	75.0	1.0	81.0	0.0	38.0	1.0	368000.0	4.0	131.0	1.0
12	62.0	0.0	231.0	0.0	25.0	1.0	253000.0	0.9	140.0	1.0
13	45.0	1.0	981.0	0.0	30.0	0.0	136000.0	1.1	137.0	1.0
14	50.0	1.0	168.0	0.0	38.0	1.0	276000.0	1.1	137.0	1.0
15	49.0	1.0	80.0	0.0	30.0	1.0	427000.0	1.0	138.0	0.0
16	82.0	1.0	379.0	0.0	50.0	0.0	47000.0	1.3	136.0	1.0
17	87.0	1.0	149.0	0.0	38.0	0.0	262000.0	0.9	140.0	1.0
18	45.0	0.0	582.0	0.0	14.0	0.0	166000.0	0.8	127.0	1.0
19	70.0	1.0	125.0	0.0	25.0	1.0	237000.0	1.0	140.0	0.0
20	48.0	1.0	582.0	1.0	55.0	0.0	87000.0	1.9	121.0	0.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	48.589028885707 15	0.0	958.0	0.0	21.0	0.0	320541.141678598 2	1.1	136.0	1.0
2	49.943642428184 29	0.0	279.0	0.0	34.0	1.0	319000.0	1.00563575718157 08	128.0	0.0
3	55.013358531571 13	1.0	110.0	0.0	25.0	0.0	406423.188443466 03	1.19866414684288 71	139.0	0.0
4	69.498368096837 07	0.0	523.0	1.0	45.0	1.0	185300.979141897 75	1.17993472387348 25	139.0	1.0
5	59.404260346691 06	1.0	278.0	0.0	54.0	0.0	442283.786818257 5	1.08014201155636 87	132.0	0.0
6	64.055508131815 69	0.0	1185.0	0.0	39.0	0.0	359322.288615458	1.77555934945474 46	141.0	0.0
7	45.0	1.0	66.0	1.0	25.0	0.0	233000.0	0.8	135.0	1.0
8	70.0	0.0	161.0	0.0	25.0	0.0	244000.0	1.2	142.0	0.0
9	58.0	1.0	400.0	0.0	40.0	0.0	164000.0	1.0	139.0	0.0
10	55.0	0.0	572.0	1.0	35.0	0.0	231000.0	0.8	143.0	0.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	82.979774848511 88	0.0	34.0	0.0	43.0	0.0	361616.180121190 5	3.20202251514881 15	131.0	1.0
2	67.925374518088 29	0.0	531.0	0.0	21.0	0.0	265731.343629522 1	1.2	134.0	1.0
3	48.2776482218726	0.0	564.0	0.0	22.0	0.0	119001.778120866 6	1.57342447387670 87	135.0	1.0
4	49.901743979665 86	0.0	602.0	0.0	36.0	0.0	310884.304183007 3	1.82139518373268 63	135.0	1.0
5	45.079289001756 32	0.0	582.0	0.0	14.0	0.0	165793.848595433 58	0.79682843992974 73	127.0	0.0
6	62.805233689733 896	0.0	361.0	0.0	26.0	0.0	266609.495294195 13	1.18536822459822 6	136.0	1.0
7	60.0	0.0	2656.0	1.0	30.0	0.0	305000.0	2.3	137.0	1.0
8	45.0	1.0	130.0	0.0	35.0	0.0	174000.0	0.8	139.0	1.0
9	45.0	0.0	582.0	1.0	55.0	0.0	543000.0	1.0	132.0	0.0
10	65.0	1.0	59.0	1.0	60.0	0.0	172000.0	0.9	137.0	0.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	85.6880970487985 8	0.0	582.0	0.0	37.0	0.0	263358.03	1.83	134.0	0.0
2	51.4437551405825 96	0.0	581.0	0.0	47.0	0.0	154042.712045976 3	0.63208344756850 22	134.0	0.0
3	51.7301508606595 3	0.0	1347.0	0.0	22.0	0.0	316724.874282783 7	1.1	136.0	1.0
4	60.0	0.0	254.0	0.0	28.0	0.0	211644.212498462 82	2.16977499956080 16	134.0	1.0
5	56.6508365756653 2	0.0	211.0	0.0	29.0	0.0	110068.897535107 52	2.62090304797211	125.0	0.0
6	71.5045127195698 7	0.0	290.0	0.0	43.0	0.0	233843.863012329 7	2.66516242681004 16	134.0	0.0
7	50.0	0.0	582.0	1.0	38.0	0.0	310000.0	1.9	135.0	1.0
8	79.0	1.0	55.0	0.0	50.0	1.0	172000.0	1.8	133.0	1.0
9	65.0	1.0	160.0	1.0	20.0	0.0	327000.0	2.7	116.0	0.0
10	70.0	0.0	92.0	0.0	60.0	1.0	317000.0	0.8	140.0	0.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	75.1776804820676 4	0.0	651.0	0.0	25.0	0.0	263358.03	1.60289903544151 76	134.0	1.0
2	75.6958572769831 6	0.0	209.0	0.0	40.0	0.0	223652.071361508 42	2.25644995305589 2	137.0	1.0
3	62.0009132319051 2	0.0	291.0	1.0	39.0	0.0	328272.395188398 14	2.81809879709953 4	142.0	0.0
4	61.6725157675142 1	1.0	249.0	0.0	33.0	0.0	157774.270497768 48	4.03719243159317 9	135.0	1.0
5	76.4174322005034	0.0	2588.0	0.0	29.0	0.0	239566.972880201 36	1.0	136.0	0.0
6	46.9464332342689 5	0.0	397.0	0.0	18.0	1.0	271000.0	1.87285602377545 28	125.0	0.0
7	48.0	1.0	131.0	1.0	30.0	1.0	244000.0	1.6	130.0	0.0
8	95.0	1.0	371.0	0.0	30.0	0.0	461000.0	2.0	132.0	1.0
9	45.0	1.0	981.0	0.0	30.0	0.0	136000.0	1.1	137.0	1.0
10	50.0	0.0	582.0	1.0	38.0	0.0	310000.0	1.9	135.0	1.0

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	64.6848063402939	1.0	246.0	1.0	25.0	0.0	197873.92253611758	1.4693426051353398	129.0	0.0
2	65.6696148806153	0.0	577.0	0.0	28.0	0.0	164316.94408044437	0.9482136640136735	137.0	0.0
3	76.74395291213244	0.0	1785.0	0.0	46.0	0.0	263358.03	3.5993290010269905	138.0	0.0
4	51.41712473478663	0.0	287.0	0.0	37.0	0.0	216531.421775545	2.1051453489668384	132.0	0.0
5	59.25812816800105	1.0	150.0	0.0	25.0	0.0	213338.423243995 3	1.5145320420002 62	135.0	1.0
6	62.44639524210376	0.0	236.0	0.0	24.0	1.0	253044.6395242104	0.9178558096841505	139.0	1.0
7	78.0	0.0	224.0	0.0	50.0	0.0	481000.0	1.4	138.0	1.0
8	70.0	0.0	582.0	0.0	20.0	1.0	263358.03	1.83	134.0	1.0
9	61.0	0.0	248.0	0.0	30.0	1.0	267000.0	0.7	136.0	1.0
10	40.0	0.0	478.0	1.0	30.0	0.0	303000.0	0.9	136.0	1.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	74.95877675233974	0.0	582.0	0.0	29.0	1.0	263358.03	1.83	134.0	0.0
2	56.42945707279832	1.0	204.0	0.0	50.0	0.0	152214.72853639917	7.05643434176134	135.0	0.0
3	60.71332722595645 5	0.0	515.0	1.0	36.0	0.0	457562.6104787994	0.7754784975852872	137.0	1.0
4	50.04555074701113 4	1.0	130.0	0.0	29.0	1.0	243349.14294413282	1.5442122523542419	130.0	0.0
5	47.12711218703093	1.0	954.0	0.0	29.0	0.0	136243.09853566068	1.3005562919200593	136.0	1.0
6	59.08357055825574	0.0	557.0	0.0	40.0	0.0	216908.35705582556	3.5716998781558043	133.0	0.0
7	81.0	0.0	4540.0	0.0	35.0	0.0	231000.0	1.18	137.0	1.0
8	70.0	0.0	2695.0	1.0	40.0	0.0	241000.0	1.0	137.0	1.0
9	70.0	0.0	835.0	0.0	35.0	1.0	305000.0	0.8	133.0	0.0
10	80.0	0.0	776.0	1.0	38.0	1.0	192000.0	1.3	135.0	0.0

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	78.759198088099	0.0	4348.0	0.0	32.0	0.0	245713.39213560915	0.9728660786439086	134.0	1.0
2	72.02847492377768	0.0	582.0	0.0	20.0	1.0	264024.16899411904	1.8583986489328874	132.0	1.0
3	67.84135436322487	0.0	207.0	0.0	22.0	0.0	95979.20224064105	1.438621521225838	132.0	0.0
4	66.06271553815498	0.0	444.0	0.0	23.0	0.0	265265.67888453876	1.2	135.0	1.0
5	63.13861899767362	0.0	277.0	1.0	53.0	0.0	190233.65720279166	1.0372276200465278	140.0	0.0
6	79.01034326927804	0.0	3597.0	0.0	31.0	0.0	243399.310448714 8	1.0798620897429594	135.0	0.0
7	75.0	0.0	582.0	1.0	30.0	1.0	263358.03	1.83	134.0	0.0
8	53.0	0.0	207.0	1.0	40.0	0.0	223000.0	1.2	130.0	0.0
9	60.0	1.0	2281.0	1.0	40.0	0.0	283000.0	1.0	141.0	0.0
10	59.0	1.0	280.0	1.0	25.0	1.0	302000.0	1.0	141.0	0.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	61.041767509914095	0.0	244.0	0.0	31.0	0.0	253958.2324900859	1.522851118555839	136.0	0.0
2	61.27686761246879	1.0	167.0	1.0	26.0	0.0	221367.2367116885	1.0	135.0	1.0
3	78.61039928730388	1.0	125.0	0.0	38.0	0.0	382858.4773630243	8.27442342271613 6	133.0	0.0
4	78.45999338729986	1.0	377.0	0.0	45.0	0.0	223336.92155091534	1.7101542530892399	143.0	1.0
5	52.6110751729081	0.0	1331.0	0.0	31.0	0.0	312088.8601383265	1.4822279723347036	136.0	1.0
6	72.92844705845354	0.0	915.0	0.0	28.0	0.0	263358.03	1.83	134.0	1.0
7	65.0	0.0	224.0	1.0	50.0	0.0	149000.0	1.3	137.0	1.0
8	80.0	0.0	898.0	0.0	25.0	0.0	149000.0	1.1	144.0	1.0
9	66.0	1.0	72.0	0.0	40.0	1.0	242000.0	1.2	134.0	1.0
10	75.0	1.0	582.0	0.0	30.0	0.0	225000.0	1.83	134.0	1.0

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	50.624824690134889	0.0	563.0	0.0	48.0	0.0	153374.89481408094	0.6291584855396283	133.0	0.0
2	45.371060303132616	0.0	917.0	0.0	30.0	0.0	137261.60503065088	1.1074212060626525	136.0	0.0
3	78.733397862257089	0.0	748.0	0.0	35.0	0.0	263358.03	1.2849239121104647	134.0	0.0
4	69.428113249408601	0.0	123.0	0.0	44.0	1.0	283771.24529976345	1.2942811324940862	136.0	1.0
5	45.0	0.0	911.0	0.0	28.0	0.0	134255.12545809412	1.1872437270952945	136.0	1.0
6	45.0	0.0	4400.0	0.0	29.0	0.0	387681.591950388 3	1.0	141.0	1.0
7	55.0	0.0	572.0	1.0	35.0	0.0	231000.0	0.8	143.0	0.0
8	60.0	0.0	68.0	0.0	20.0	0.0	119000.0	2.9	127.0	1.0
9	70.0	0.0	97.0	0.0	60.0	1.0	220000.0	0.9	138.0	1.0
10	53.0	1.0	1808.0	0.0	60.0	1.0	249000.0	0.7	138.0	1.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	57.170521424634856	1.0	442.0	0.0	63.0	1.0	149658.625551116	8.439061594103064	136.0	0.0
2	80.070226985342309	0.0	803.0	0.0	38.0	0.0	263358.03	1.1036618356642813	134.0	1.0
3	63.119061490902704	0.0	131.0	0.0	28.0	0.0	220505.60632085375	1.5914576434219743	136.0	1.0
4	57.072892954324644	1.0	97.0	0.0	25.0	0.0	402594.50949844596	1.4531223810398415	137.0	0.0
5	46.998360467488595	0.0	165.0	0.0	17.0	0.0	271115.1954385564	2.0577616725293279	124.0	0.0
6	60.5750453439411	0.0	351.0	1.0	29.0	0.0	385691.2867385102	1.5827178315372452	137.0	0.0
7	60.0	0.0	897.0	1.0	45.0	0.0	297000.0	1.0	133.0	1.0
8	75.0	0.0	582.0	0.0	20.0	1.0	265000.0	1.9	130.0	1.0
9	45.0	1.0	1876.0	1.0	35.0	0.0	226000.0	0.9	138.0	1.0
10	75.0	1.0	246.0	0.0	15.0	0.0	127000.0	1.2	137.0	1.0